

Previous Year Question Paper

of

G.A.T.E. (BT) 2013

BIOTECHNOLOGY

Examination

(Original Question Paper with Answer Key)

GRADUATE APTITUDE TEST IN ENGINEERING





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BT:BIOTECHNOLOGY

Duration: Three Hours *Maximum Marks*:100

Please read the following instructions carefully:

General Instructions:

1. Total duration of examination is 180 minutes (3 hours).
2. The clock will be set at the server. The countdown timer in the top right corner of screen will display the remaining time available for you to complete the examination. When the timer reaches zero, the examination will end by itself. You will not be required to end or submit your examination.
3. The Question Palette displayed on the right side of screen will show the status of each question using one of the following symbols:



You have not visited the question yet.

You have not answered the question.

You have answered the question.

You have NOT answered the question, but have marked the question for review.

You have answered the question, but marked it for review.

The Marked for Review status for a question simply indicates that you would like to look at that question again. ***If a question is answered and Marked for Review, your answer for that question will be considered in the evaluation.***

Navigating to a Question

4. To answer a question, do the following:
   a. Click on the question number in the Question Palette to go to that question directly.
   b. Select an answer for a multiple choice type question. Use the virtual numeric keypad to enter a number as answer for a numerical type question.
   c. Click on **Save and Next** to save your answer for the current question and then go to the next question.
   d. Click on **Mark for Review and Next** to save your answer for the current question, mark it for review, and then go to the next question.
   e. **Caution: Note that your answer for the current question will not be saved, if you navigate to another question directly by clicking on its question number.**
5. You can view all the questions by clicking on the **Question Paper** button. Note that the options for multiple choice type questions will not be shown.

Answering a Question

6. Procedure for answering a multiple choice type question:
    a. To select your answer, click on the button of one of the options
    b. To deselect your chosen answer, click on the button of the chosen option again or click on the **Clear Response** button
    c. To change your chosen answer, click on the button of another option
    d. To save your answer, you MUST click on the **Save and Next** button
    e. To mark the question for review, click on the **Mark for Review and Next** button**.** ***If an answer is selected for a question that is Marked for Review, that answer will be considered in the evaluation.***

7. Procedure for answering a numerical answer type question:
    a. To enter a number as your answer, use the virtual numerical keypad
    b. A fraction (eg.,-0.3 or -.3) can be entered as an answer with or without '0' before the decimal point
    c. To clear your answer, click on the **Clear Response** button
    d. To save your answer, you MUST click on the **Save and Next** button
    e. To mark the question for review, click on the **Mark for Review and Next** button**.** ***If an answer is entered for a question that is Marked for Review, that answer will be considered in the evaluation.***

8. To change your answer to a question that has already been answered, first select that question for answering and then follow the procedure for answering that type of question.

9. Note that ONLY Questions for which answers are saved or marked for review after answering will be considered for evaluation.

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Paper specific instructions:

1. There are a total of 65 questions carrying 100 marks. Questions are of multiple choice type or numerical answer type. A multiple choice type question will have four choices for the answer with only **one** correct choice. For numerical answer type questions, the answer is a number and no choices will be given**. A number as the answer should be entered** using the virtual keyboard on the monitor.
2. Questions Q.1 – Q.25 carry 1mark each. Questions Q.26 – Q.55 carry 2marks each. The 2marks questions include two pairs of common data questions and two pairs of linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is not attempted, then the answer to the second question in the pair will not be evaluated.
3. Questions Q.56 – Q.65 belong to General Aptitude (GA) section and carry a total of 15 marks. Questions Q.56 – Q.60 carry 1mark each, and questions Q.61 – Q.65 carry 2marks each.
4. Questions not attempted will result in zero mark. Wrong answers for multiple choice type questions will result in **NEGATIVE** marks. For all 1 mark questions, ⅓ mark will be deducted for each wrong answer. For all 2 marks questions, ⅔ mark will be deducted for each wrong answer. However, in the case of the linked answer question pair, there will be negative marks only for wrong answer to the first question and no negative marks for wrong answer to the second question. There is no negative marking for questions of numerical answer type.
5. Calculator is allowed. Charts, graph sheets or tables are **NOT** allowed in the examination hall.
6. Do the rough work in the Scribble Pad provided.

Q. 1 – Q. 25 carry one mark each.

Q.1 Under alkaline conditions, DNA is more stable than RNA because

(A) RNA forms secondary structures
(B) RNA is a single stranded molecule
(C)RNA has uracil in place of thymidine
(D)RNA is susceptible to hydrolysis

Ans. D

Q.2 Which one of the following modifications is common toboth protein and DNA?

(A) SUMOylation (B) Nitrosylation (C)Methylation (D) Ubiquitination

Ans. C

Q.3 Protein A, which has strong affinity to F_c region of immunoglobulin, is extracted from

(A)*Saccharomyces cerevisiae*
(B)*Staphylococcus aureus*
(C)*Streptococcus pyogenes*
(D)*Streptococcus sanguis*

Ans. B

Q.4 The first humanized monoclonal antibody approved for the treatment of breast canceris

(A) Rituximab (B) Cetuximab (C) Bevacizumab (D) Herceptin

Ans. D

Q.5 Which one of the following aminoacids in proteins does **NOT** undergo phosphorylation?

(A) Ser (B) Thr (C) Pro (D) Tyr

Ans. C

Q.6 The role of an adjuvant isto

(A) prolong the persistence of antigen
(B) cross link the antigen
(C) increase the size of antigen
(D) avoid inflammation

Ans. A

Q.7 Endogenous antigens are presented on to the cell surface along with

(A) MHC-II (B) MHC-I (C) $F_c\gamma$receptor (D) complement receptor

Ans. B

Q.8 Human genome sequencing project involved the construction of genomic library in

(A)bacterial artificial chromosome (B)pBR322
(C) bacteriophage (D) pcDNA3.1

Ans. A

Q.9 The nucleotide analogue used in DNA sequencing by chain termination method is

(A) 1',3'-dideoxy nucleoside triphosphate
(B) 2',3'-dideoxy nucleoside triphosphate
(C) 2',4'-dideoxy nucleoside triphosphate
(D) 2',5'-dideoxy nucleoside triphosphate

Ans. B

Q.10 In nature, the horizontal gene transfer across bacteria is mediated by

(A) gene cloning followed by transformation (B)conjugation and transformation
(C) conjugation only (D) transformation only

Ans. B

Q.11 Phylum proteobacteria is subdivided into α-, β-, γ-, δ- and ε-proteobacteria based on

(A) G+C content (B) 23S rRNA sequences
(C)tRNA sequences (D)16S rRNA sequences

Ans. D

EBC Print less... Save paper... Save trees...

Q.12 Which one of the following is an ABC transporter?

(A)multidrug resistance protein (B) acetylcholine receptor
(C)bacteriorhodopsin (D) ATP synthase

Ans. A

Q.13 The catalytic efficiency for an enzyme is definedas

(A) k_{cat} (B) $\frac{V_{max}}{k_{cat}}$ (C) $\frac{k_{cat}}{K_m}$ (D) $\frac{k_{cat}}{V_{max}}$

Ans. C

Q.14 Of the two diploid species, species I has 36 chromosomes and species II has 28 chromosomes. How many chromosomes would be found in an allotriploid individual?

(A) 42 or 54 (B)46 or 50 (C) 74 or 86 (D) 84 or 108

Ans. B

Q.15 The RNA primer synthesized during the replication process in bacteria is removed by

(A) DNA gyrase (B)primase
(C) DNA polymeraseI (D) DNA polymerase II

Ans. C

Q.16 The suitable substitution matrix to align closely related sequences is

(A) PAM 250 or BLOSSUM 80 (B) PAM 40 or BLOSSUM 80
(C) PAM 120 or BLOSSUM 40 (D) PAM 250 or BLOSSUM 40

Ans. B

Q.17 If $P=\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}, Q=\begin{bmatrix}2 & 1\\ 2 & 2\end{bmatrix}$ and $R=\begin{bmatrix}3 & 0\\ 1 & 3\end{bmatrix}$,which one of the following statements is **TRUE**?

(A) PQ = PR (B) QR= RP
(C) QP = RP (D) PQ = QR

Ans. A

Q.18 If u = log $(e^x + e^y)$, then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$=

(A) $e^x + e^y$ (B) $e^x - e^y$
(C) $\frac{1}{e^x + e^y}$ (D) 1

Ans. D

Q.19 Hypophosphatemia is manifested by an X-linked dominant allele.What proportion of the offsprings from a normal male and an affected heterozygous female will manifest the disease?

(A) ½ sons and ½ daughters (B) all daughters and no sons
(C) all sons and no daughters (D) ¼ daughters and ¼ sons

Ans. A

Q.20 One of theeigen values of $P=\begin{bmatrix}10 & -4\\ 18 & -12\end{bmatrix}$ is

(A) 2 (B) 4 (C) 6 (D) 8

Ans. C

Q. 21 – Q. 25 are of numerical answer type.

Q.21 A callus of 5 g dry weight was inoculated on semi-solid medium for growth. The dry weight of the callus was found to increase by 1.5 fold after 10 days of inoculation. The growth index of the culture is _____

Ans. 0.5

Q.22 A chemostat is operated at a dilution rate of 0.6 h^{-1}. At steady state, the biomass concentration in the exit stream was found to be 30 g l^{-1}. The biomass productivity (g $l^{-1}h^{-1}$) after 3h of steady state operation will be _____

Ans. 18

Q.23 A batch bioreactor is to be scaled up from 10 to 10,000 liters. The diameter of the large bioreactor is 10 times that of the small bioreactor. The agitator speed in the small bioreactor is 450 rpm. Determine the agitator speed (rpm) of the large bioreactor with same impeller tip speed as that of the small bioreactor. _____

Ans. 45

Q.24 Calculate the percentage sequence identity for the pairwise alignment given below. _____

H E L L O –
Y E L L O W

Ans. 66.5 to 66.7

Q.25 In a batch culture, the specific rate of substrate utilization is 0.25 g (g cell mass)$^{-1}$ h^{-1} and specific rate of product formation is 0.215 g (g cell mass)$^{-1}$ h^{-1}. Calculate the yield of product from the substrate($Y_{p/s}$). _____

Ans. 0.81 to 0.91

Q. 26 to Q. 55 carry two marks each.

Q.26 Match the commercial microbial sources in **Group I** with the products in **Group II.**

Group I	**Group II**
P. *Corynebacteriumlilium*	1. 2,3-Butane di-ol
Q. *Klebsiellaoxytoca*	2. Poly-β-hydroxybutyric acid
R. *Aspergillusniger*	3. Glutamic acid
S. *Alcaligeneseutrophus*	4. Citric acid

(A) P-3,Q-1,R-2,S-4 (B) P-3,Q-1,R-4,S-2
(C) P-1,Q-3,R-2,S-4 (D) P-1,Q-3,R-4,S-2

Ans. B

Q.27 Match the entries in the **Group I** with the elution conditions in **Group II.**

Group I	**Group II**
P. Ion-exchange chromatography	1. Isocratic solvent
Q. Hydrophobic column chromatography	2. Ampholytes
R. Gel filtration chromatography	3. Increasing gradient of salt
S. Chromatofocusing	4. Decreasing gradient of polarity

(A) P-4,Q-1,R-2,S-3 (B) P-4,Q-3,R-1,S-2
(C) P-3,Q-4,R-1,S-2 (D) P-3,Q-4,R-2,S-1

Ans. C

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Q.28 Determine the correctness or otherwise of the following **Assertion** (a) and **Reason** (r).

Assertion: Immobilization of plant cells can enhance secondary metabolite production during bioreactor cultivation.

Reason: Immobilization protects the plant cells from shear forces in the bioreactor.

(A) Both (a) and (r) are true and (r) is the correct reason for (a).
(B) Both (a) and (r) are true but (r) is not the correct reason for (a).
(C) (a) is true but (r) is false.
(D) (a) is false but (r) is true.

Ans. A

Q.29 Match the cell structures in **Group I** with the organismsin **Group II.**

Group I	**Group II**
P. Endospores	1. *Methanobacterium*
Q. Bipolar flagella	2. *Treponema*
R. Pseudomurine in cell wall	3. *Spirillum*
S. Periplasmic flagella	4. *Clostridium*

(A) P-4, Q-3, R-1, S-2 (B) P-4, Q-3, R-2, S-1
(C) P-3, Q-4, R-1, S-2 (D) P-4, Q-1, R-3, S-2

Ans. A

Q.30 Match the antibioticsin**Group I**with the targets in**Group II.**

Group I	**Group II**
P. Sulfonamide	1. Peptidoglycan synthesis
Q. Quinolones	2. Peptide chain elongation
R. Erythromycin	3. Folic acid biosynthesis
S. Cephalosporin	4. Topoisomerase

(A) P-3, Q-4, R-1, S-2 (B) P-2, Q-4, R-3, S-1
(C) P-4, Q-1, R-2, S-3 (D) P-3, Q-4, R-2, S-1

Ans. D

Q.31 In nature, *Agrobacterium tumefaciens* mediated infection of plant cells leads to

P. crown gall disease in plants
Q. hairy root disease in plants
R. transfer of T-DNA into the plant chromosome
S. transfer of Ri-plasmid into the plant cell

(A) S only (B) P and R only (C) Q and S only (D) Q only

Ans. B

Q.32 Match the entries in**Group I** with the enzymes in**Group II.**

Group I	**Group II**
P. NAD^+	1. Glutathione peroxidase
Q. Selenium	2. Nitrogenase
R. Pyridoxal phosphate	3. Lactate dehydrogenase
S. Molybdenum	4. Glycogen phosphorylase

(A) P-3, Q-2, R-4, S-1 (B) P-4, Q-1, R-3, S-2
(C) P-3, Q-1, R-4, S-2 (D) P-3, Q-4, R-2, S-1

Ans. C

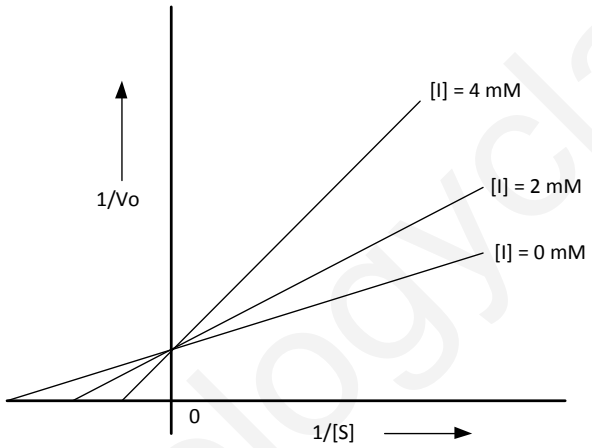
Q.33 Match the herbicides in **Group I**with the target enzymesin **Group II.**

Group I	**Group II**
P. Glyphosate	1. Nitrilase
Q. Bromoxynil	2. Acetolactatesynthetase
R. Sulphonylureas	3. Dehalogenase
S. Dalapon	4. 5-Enolpyruvyl shikimate3-phosphate synthase

(A) P-4, Q-1, R-2, S-3
(B) P-2, Q-1, R-4, S-3
(C) P-4, Q-3, R-2, S-1
(D) P-3, Q-2, R-4, S-1

Ans. A

Q.34 The activity of an enzyme was measured by varying the concentration of the substrate (S) in the presence of three different concentrations of inhibitor (I) 0, 2 and 4 mM.The double reciprocal plot given belowsuggests that the inhibitor (I) exhibits



(A) substrate inhibition
(B) uncompetitive inhibition
(C) mixed inhibition
(D) competitive inhibition

Ans. D

Q.35 Match the entries in **Group I**with the entries in **Group II.**

Group I	**Group II**
P. RNAse P	1. Polyadenylation
Q. RNase H	2. Splicing
R. snRNAs	3. Ribozymes
S. CstF	4. DNA-RNAhybrids

(A) P-3, Q-4, R-2, S-1
(B) P-4, Q-3, R-2, S-1
(C) P-3, Q-2, R-1, S-4
(D) P-2, Q-4, R-1, S-3

Ans. A

Q.36 Determine the correctness or otherwise of the following **Assertion** (a) and **Reason** (r).

Assertion:UPGMA method produces ultrametric tree.

Reason: Sequence alignment is converted into evolutionary distances in UPGMA method.

(A) Both (a) and (r) are true and (r) is the correct reason for (a)
(B) Both (a) and (r) are true and (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Ans. B

Q.37 Match the entries in the **Group I**with the entries in**Group II.**

Group I	**Group II**
P. Threading	1. Gene duplication
Q. FASTA	2. Fold prediction
R. Profile	3. HMM
S. Paralogs	4. k-tuple

(A) P-2, Q-1, R-3, S-4 (B) P-2, Q-4, R-3, S-1
(C) P-3, Q-4, R-2, S-1 (D) P-1, Q-4, R-3, S-2

Ans. B

Q.38 Evaluate $\lim_{x\to\infty} x \tan\frac{1}{x}$

(A) ∞ (B) 1 (C) 0 (D)-1

Ans. B

Q.39 The Laplace transform of $f(t) = 2t + 6$ is

(A) $\frac{1}{s} + \frac{2}{s^2}$ (B) $\frac{3}{s} - \frac{6}{s^2}$ (C) $\frac{6}{s} + \frac{2}{s^2}$ (D) $-\frac{6}{s} + \frac{2}{s^2}$

Ans. C

Q.40 The solution of the following set of equations is

$$x + 2y + 3z = 20$$
$$7x + 3y + z = 13$$
$$x + 6y + 2z = 0$$

(A) $x = -2,\ y = 2,\ z = 8$ (B) $x = 2,\ y = -3,\ z = 8$
(C) $x = 2,\ y = 3,\ z = -8$ (D) $x = 8,\ y = 2,\ z = -3$

Ans. B

Q.41 The solution to $\frac{dy}{dx} + y\ cot\ x = cosec\ x$ is

(A) $y = (c+x)\ cot\ x$ (B) $y = (c+x)\ cosec\ x$
(C) $y = (c + x)\ cosec\ x\ cot\ x$ (D) $y = (c+x)\frac{cosec\ x}{\cot x}$

Ans. B

Q.42 A complete restriction digestion of a circular plasmid (5000bp) was carried out with *Hin*dIII,*Bam*HIand*Eco*RIindividually. Restriction digestion yielded following fragments.

Plasmid + *Hin*dIII→ 1200bp and3800bp
Plasmid + *Bam*HI→5000bp
Plasmid + *Eco*RI→ 2500bp

Ans. A

The number of sites for *Eco*RI,*Bam*HI*andHin*dIIIpresent on this plasmid are

(A)*Eco*RI-2, *Bam*HI-1, *Hin*dIII-2 (B)*Eco*RI-1, *Bam*HI-1, *Hin*dIII-2
(C)*Eco*RI-3, *Bam*HI-2, *Hin*dIII-1 (D)*Eco*RI-2, *Bam*HI-2, *Hin*dIII-1

Q. 43 – Q. 47 are of numerical answer type.

Q.43 The total number of fragments generated by the complete and sequential cleavage of the polypeptide given below by Trypsin followed by CNBr is _____

Phe-Trp-Met-Gly-Ala-Lys-Leu-Pro-Met-Asp-Gly-Arg-Cys-Ala-Gln

Ans. 5

Q.44 In a genetic study, 80 people were found to have alleles for polydactyly. Only 36 of them were polydactylous. What is the extent of penetrance percentage? _____

Ans. 45

Q.45 One percent of the cars manufactured by a company are defective. What is the probability (upto four decimals) that more than two cars are defective, if 100 cars are produced? _____

Ans. 0.075 to 0.085

Q.46 The maximum cell concentration (g l^{-1}) expected in a bioreactor with initial cell concentration of 1.75 g l^{-1} and an initial glucose concentration of 125 g l^{-1} is ($Y_{x/s}$ = 0.6 g cell/g substrate) _____

Ans. 76.7 to 76.9

Q.47 A fed batch culture was operated with intermittent addition of glucose solution at a flow rate of 200 ml h^{-1}. The values of K_s, μ_m and D are 0.3 g l^{-1}, 0.4 h^{-1} and 0.1 h^{-1}, respectively. Determine the concentration of growth limiting substrate (gl^{-1}) in the reactor at quasi-steady state. _____

Ans. 0.1

Common Data Questions

Common Data for Questions 48 and 49:

A solution was prepared by dissolving 100 mg of protein X in 100 ml of water.
Molecular weight of protein X is 15,000 Da; Avogadro's number = 6.022×10^{23}.

Q.48 Calculate the molarity(μM) of the resulting solution.

(A)66.6 (B)6.6
(C)0.67 (D)0.067

Ans. A

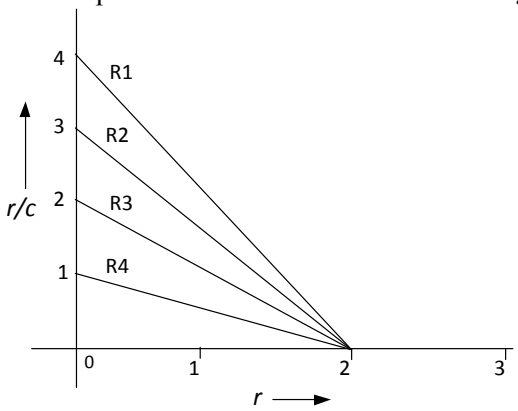
Q.49 The number of moleculespresent in this solution is

(A) 40.15×10^{19} (B) 6.023×10^{19} (C)4.015×10^{19} (D) 0.08×10^{19}

Marks to all

Common Data for Questions 50 and 51:

The binding efficiency of three different receptorsR1, R2 and R3 were tested against a ligand using equilibrium dialysis, with a constant concentration of receptor and varying concentrations of ligand. The Scatchard plot of receptor titration with different concentration of ligand is given below
(*r*is moles of bound ligand per moles of receptor and *c* is concentration of free ligand)



Q.50 The number of ligand binding sites present on receptors R1 and R3, respectively are

(A)1 and 4 (B)1 and 1 (C)4 and 1 (D)2 and 2

Ans. D

Q.51 Which one of the receptors has thehighestaffinity for the ligand?

(A) R1 (B) R2 (C) R3 (D) R4

Ans. a

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

A DNA fragment of 5000bp needs to be isolated from *E.coli* (genome size $4x10^3$kb) genomic library.

Q.52 The minimum number of independent recombinant clones required to represent this fragment in genomic library are

(A) $16x10^2$ (B)$12x10^2$ (C) $8x10^2$ (D) $1.25x10^2$ *Ans. C*

Q.53 The number of clones to represent this fragment in genomic library with a probability of 95% are

(A) $5.9\ x10^3$ (B) $4.5x10^3$ (C) $3.6\ x10^3$ (D) $2.4x10^3$

Ans. D

Statement for Linked Answer Questions 54 and 55:

During sterilization of a fermentation medium in a given bioreactor$\nabla_{heating}$= 12.56, $\nabla_{cooling}$ = 7.48 and the total value of ∇ required for whole sterilization process is 52, where ∇ is the design criteria.

Q.54 What is the value of $\nabla_{holding}$?

(A)31.96 (B) 42.32 (C) 52.43 (D) 61.18

Ans. A

Q.55 What is the holding period (min) at a k value of $3.36min^{-1}$?

(A) 10.6 (B) 9.5 (C) 8.4 (D) 7.2

Ans. B

General Aptitude (GA) Questions

Q. 56 – Q. 60 carry one mark each.

Q.56 If $3 \leq X \leq 5$ and $8 \leq Y \leq 11$ then which of the following options is TRUE?

(A) $\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{5}$

(B) $\frac{3}{11} \leq \frac{X}{Y} \leq \frac{5}{8}$

(C) $\frac{3}{11} \leq \frac{X}{Y} \leq \frac{8}{5}$

(D) $\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{11}$

Ans. B

Q.57 The Headmaster _____ to speak to you.

Which of the following options is incorrect to complete the above sentence?

(A) is wanting

(B) wants

(C) want

(D) was wanting

Ans. C

Q.58 Mahatama Gandhi was known for his humility as

(A) he played an important role in humiliating exit of British from India.

(B) he worked for humanitarian causes.

(C) he displayed modesty in his interactions.

(D) he was a fine human being.

Ans. C

Q.59 All engineering students should learn mechanics, mathematics and how to do computation.
I II III IV

Which of the above underlined parts of the sentence is not appropriate?

(A) I (B) II (C) III (D) IV

Ans. D

Q.60 Select the pair that best expresses a relationship similar to that expressed in the pair:
water: pipe::

(A) cart: road
(B) electricity: wire
(C) sea: beach
(D) music: instrument

Ans. B

Q. 61 to Q. 65 carry two marks each.

Q.61 Velocity of an object fired directly in upward direction is given by $V = 80 - 32\,t$, where t (time) is in seconds. When will the velocity be between 32 m/sec and 64 m/sec?

(A) (1, 3/2) (B) (1/2, 1)

(C) (1/2, 3/2) (D) (1, 3)

Ans. C

Q.62 In a factory, two machines M1 and M2 manufacture 60% and 40% of the autocomponents respectively. Out of the total production, 2% of M1 and 3% of M2 are found to be defective. If a randomly drawn autocomponent from the combined lot is found defective, what is the probability that it was manufactured by M2?

(A) 0.35 (B) 0.45 (C) 0.5 (D) 0.4

Ans. C

Q.63 Following table gives data on tourists from different countries visiting India in the year 2011.

Country	Number of Tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

Which two countries contributed to the one third of the total number of tourists who visited India in 2011?

(A) USA and Japan
(B) USA and Australia
(C) England and France
(D) Japan and Australia

Ans. C

Q.64 If $|-2X + 9| = 3$ then the possible value of $|-X| - X^2$ would be:

(A) 30 (B) -30 (C) -42 (D) 42

Ans. B

Q.65 All professors are researchers
Some scientists are professors

Which of the given conclusions is logically valid and is inferred from the above arguments:

(A) All scientists are researchers
(B) All professors are scientists
(C) Some researchers are scientists
(D) No conclusion follows

Ans. C

END OF THE QUESTION PAPER

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